

Weekly Progress Report

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Domain: **Data Science and Machine Learning**

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Week Ending: 03

I. Overview:

This week, the primary focus transitioned from initial baseline exploration to designing and developing an application for the project Quality Prediction in a Mining Process. Efforts centred on converting the developed interface into a user-friendly application that calculates and visualizes mining quality predictions based on key process inputs.

II. Achievements:

1. Application Development

Designed and developed an interactive application interface using Streamlit. Developed input fields such as sliders and numerical inputs for critical engineering features: pH levels, slurry intensity, and chemical reagent dosing rates. Created reactive application elements that dynamically display predicted mining quality variations based on user input adjustments.

2. Model Feature Integration

Restricted the model inference pipeline to focus tightly on the most impactful features identified during exploratory data analysis: pH, slurry intensity, and reagent dose. Standardized the app's internal pipeline to correctly format user interface inputs before passing them to the trained machine learning backend.

3. Data Visualization & UI Enhancements

Implemented real-time plotting using Matplotlib and Seaborn directly within the application interface to track historical vs. predicted quality deviations. Streamlined the web page layout for industrial operators, ensuring quick visibility of critical quality metrics.

III. Challenges:

1. Session State and UI Lag

Encountered performance bottlenecks and widget state loss when dynamically refreshing charts based on concurrent changes in slurry intensity and dosing sliders. Troubleshooting optimal uses of [*st.cache_resource*](#) and [*st.session_state*](#) to prevent unnecessary model re-loading during user interactions.

2. Out-of-Bounds Input Handling

Faced difficulties mapping highly skewed historical operational ranges (caused by sensor anomalies and plant downtime) into realistic UI bounding constraints for the sliders.

IV. Learning Resources:

1. Application Development Documentation

Utilized the official Streamlit documentation covering layout, widget, and caching guides to resolve session reactivity issues. Reviewed open-source industrial dashboard templates on GitHub for UI design best practices.

2. Advanced Feature Engineering & Deployment

Explored advanced tutorials on embedding pre-trained Scikit-Learn models ([*pickle/joblib*](#) file tracking) into runtime application architectures.

V. Next Week's Goals:

1. Real World Application Improvement

Next week, I will focus on improving the application so that it addresses real-world mining process challenges more effectively. I will refine the interface based on practical operator needs, improve the usability of the input controls, and ensure that the application provides clear and actionable quality predictions under realistic operating conditions.

2. Logic and Validation Pipeline

I will strengthen the core calculation and validation logic by testing the application with realistic operational scenarios and comparing its predictions against baseline industrial parameters. I will also improve input validation, handle abnormal or out-of-range values, and evaluate the application's performance using representative real-world data.

VI. Additional Comments:

- Preliminary performance tests confirm that tracking user modifications specifically to slurry intensity and chemical reagent dosing rates inside the app provides the most immediate, interpretable feedback regarding final quality metrics..